

Lab Experiment 7: Create, Delete, and Modify Local User Accounts

Objective

To learn how to **create**, **delete**, and **modify** local user accounts in Linux using basic administrative commands.

Background

In Linux, every person who uses the computer is treated as a **user**. Each user has a:

- Username
- Home directory
- Password
- Unique User ID (UID)

System administrators often need to manage these user accounts — creating new users, updating existing users, or removing old ones. Linux provides simple commands like `useradd`, `usermod`, and `userdel` to perform these tasks.

Difference Between Local and System Users

Type	Description	Example
Local User	A user created manually on a specific system.	<code>student1</code> , <code>admin</code> , <code>teacher</code>
System User	Automatically created by the operating system to run services.	<code>daemon</code> , <code>mysql</code> , <code>syslog</code>

In this experiment, we work only with **local users**.

Commands Used

Command	Purpose
<code>useradd</code>	Create a new user
<code>passwd</code>	Set or change a user's password
<code>usermod</code>	Modify existing user account details
<code>userdel</code>	Delete a user account

Steps to Perform in the Lab

Step 1: Check Current User

Display your current login name:

```
whoami
```

View user details (user ID, group ID, etc.):

```
id
```

Step 2: Create a New User

Create a new user named student1:

```
sudo useradd -m student1
```

`-m` → Creates a home directory `/home/student1`.

Set a password for this user:

```
sudo passwd student1
```

Check if the user has been created:

```
grep student1 /etc/passwd
```

Step 3: Modify an Existing User

The `usermod` command is used to make changes to an existing user account.

1. Change Username

```
sudo usermod -l studentA student1
```

(Now the username becomes `studentA`.)

2. Change Home Directory

```
sudo usermod -d /home/studentA -m studentA
```

3. Change User Password

```
sudo passwd studentA
```

Verify the modification:

```
grep studentA /etc/passwd
```

Step 4: Delete a User

Remove a user (without deleting home directory):

```
sudo userdel studentA
```

Remove a user along with their home directory:

```
sudo userdel -r studentA
```

Check if the user is removed:

```
grep studentA /etc/passwd
```

Outcome

After completing this lab, students will be able to:

- Create new local user accounts.
- Modify existing user accounts using **usermod**.
- Delete user accounts safely using **userdel**.
- Understand the structure and storage of user data in Linux.

Viva Questions

1. What command is used to create a new user in Linux?
2. What does the **-m** option in **useradd** do?
3. How can you set a password for a user?
4. What is the use of the **usermod** command?
5. How do you delete a user and their home directory together?
6. Where is user information stored in Linux?
7. What is the difference between **useradd** and **usermod**?
8. Can you change a username in Linux? Which command is used?
9. How do you verify whether a user exists?
10. What is the difference between a local user and a system user?